

Chapter 10 CRT

1. What is a GUI?

A GUI (Graphical User Interface) is a way for a user to interact with a computer program using graphical components such as buttons, text fields, labels, menus, and windows instead of typing commands.

2. Explain how code is executed in an event-driven application.

In an event-driven application, the program waits for the user to perform an action, such as clicking a button or typing text. When an event occurs, the program runs the code associated with that event, usually through an event listener.

3. Can components be added directly to a frame? Explain.

Yes, but normally components are added to a panel, and the panel is then added to the frame. A panel acts as a container that makes it easier to organize and manage the components in the GUI.

4. Can a label respond to events? Explain.

Yes. A label can technically have event listeners and respond to certain events. However, labels are normally used to display information and are not designed for user interaction like buttons or text fields.

5. Why do you think a GUI needs to be run from an event-dispatching thread?

The Event Dispatch Thread (EDT) manages GUI events and updates. Running the GUI on the EDT helps ensure that user interactions and changes to GUI components happen safely and in the correct order.

6. What is the difference between a label and a button?

A label is mainly used to display text or information to the user, while a button is designed to be clicked and perform an action. For example, "Enter your grade:" could be a label, while "SUBMIT" could be a button.

8. List three ways to control the layout of a content pane.

Three ways are:

- Layout managers — such as BorderLayout, FlowLayout, and GridLayout.
- Absolute positioning — using `setLayout(null)` and `setBounds()` to manually position components.
- Nested panels — using multiple panels with different layout managers to organize components.

13. What must first be done with numeric data typed in a text field before it can be used in a calculation?

The numeric data must be converted from a String into a numeric data type, such as an int or double.

14. What is the value of num1 in the last statement below?

The value of num1 is 8.0, because num2 is 3 and num3 is converted from the String "5" into the number 5. The calculation is $3 + 5 = 8.0$.